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Enhanced cardiovascular disease risk prediction using explainable machine learning and data balancing

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Abstract

Cardiovascular disease (CVD) remains a leading cause of mortality worldwide, making accurate and early risk prediction essential for effective preventive care. However, traditional risk assessment models often fail to capture complex nonlinear clinical patterns, suffer from class imbalance, and provide limited interpretability, restricting their clinical adoption. To address these challenges, this study proposes an explainable and balanced ensemble machine learning (ML) framework for robust CVD risk prediction. Four ensemble classifiers—Gradient Boosting (GB), Extreme Gradient Boosting (XGB), CatBoost (CB), and Extra Trees (ET)—were evaluated on two publicly available benchmark datasets: Heart Disease Classification (HDC) and Cardiovascular Disease (CD). Class imbalance was mitigated using the Synthetic Minority Over-sampling Technique (SMOTE), and model performance was optimized through systematic hyperparameter tuning. A comprehensive evaluation was conducted using 5-fold cross-validation with multiple metrics, including accuracy, recall, F1-score, and ROC-AUC. To enhance transparency, explainable artificial intelligence (XAI) techniques—SHAP, LIME, and permutation-based feature importance—were employed to analyze both global and local feature contributions. Experimental results demonstrate that CatBoost consistently outperformed competing models, achieving an accuracy of 98.88% (HDC) and 99.44% (CD), with corresponding ROC-AUC scores of 99.90% and 99.97%. SMOTE significantly improved minority-class sensitivity, while XAI analysis identified clinically relevant predictors such as troponin, kcm, glucose levels, slope, and exercise-induced angina, aligning with established cardiovascular risk factors. Overall, the proposed SMOTE–CatBoost framework delivers accurate, fair, and interpretable CVD risk prediction, offering strong potential for clinical decision support and precision medicine. Future work will focus on validating the framework using multimodal and real-world longitudinal patient data to further enhance generalizability.

Keywords Cardiovascular disease, Machine learning, Ensemble learning, CatBoost, SMOTE, Explainable artificial intelligence, Risk prediction, Precision medicine



1 Introduction

Cardiovascular diseases (CVDs) remain among the leading causes of mortality worldwide and continue to impose a substantial burden on healthcare systems [1, 2]. Despite advances in clinical management, timely identification of individuals at high cardiovascular risk remains challenging, particularly in low- and middle-income regions where access to advanced diagnostic resources is limited [3]. Consequently, accurate and early risk prediction is critical for enabling preventive intervention, improving patient outcomes, and reducing long-term healthcare costs.

CVDs arise from a complex interplay of demographic, clinical, and lifestyle-related factors, including age, hypertension, diabetes, obesity, dyslipidemia, smoking, and physical inactivity [4]. Conventional risk assessment tools, such as the Framingham Risk Score and SCORE, are widely used in clinical practice; however, they rely on a limited set of predefined variables and assume linear relationships among risk factors [5]. As a result, these models often struggle to capture nonlinear interactions among comorbidities and subclinical indicators, leading to reduced generalizability across heterogeneous populations [6].

Recent advances in machine learning (ML) have introduced powerful data-driven alternatives for cardiovascular risk prediction [7, 8]. ML models are capable of learning complex, nonlinear patterns from high-dimensional clinical data and have demonstrated superior predictive performance compared to traditional statistical approaches in multiple healthcare applications [9, 10]. In cardiovascular medicine, ensemble-based ML techniques—such as gradient boosting, random forests, and extreme gradient boosting—have shown improved accuracy in predicting major adverse cardiovascular events compared to logistic regression and rule-based models [11, 12].

Despite these advances, two critical challenges continue to limit the clinical adoption of ML-based CVD prediction models. First, most cardiovascular datasets are inherently imbalanced, with significantly fewer disease cases than non-disease cases. This imbalance biases learning algorithms toward the majority class, often resulting in high overall accuracy but poor sensitivity for high-risk patients [13, 14]. Second, many high-performing ML models lack interpretability, operating as black-box systems that provide limited insight into their decision-making processes and thereby reducing clinical trust and usability [15].

To address class imbalance, data resampling techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) and its variants have been widely adopted to enhance minority-class representation and improve sensitivity [16]. However, existing studies often apply oversampling without systematically integrating ensemble learning and explainable artificial intelligence (XAI), limiting both generalization performance and clinical transparency [17, 18].

1.1 Problem statement

Accurate and early prediction of cardiovascular disease remains a fundamental challenge due to the complex, nonlinear interactions among demographic, clinical, and lifestyle-related risk factors. Traditional risk assessment methods rely on limited feature sets and linear assumptions, which often lead to suboptimal predictive performance and reduced robustness across diverse patient populations.

Although ML-based approaches have demonstrated improved predictive capability, their practical deployment is hindered by two major limitations: (i) severe class imbalance in cardiovascular datasets, which biases models toward the majority (non-disease) class and reduces sensitivity for high-risk patients, and (ii) insufficient interpretability of many high-performing models, limiting clinical trust and adoption. These challenges highlight the need for a robust, balanced, and interpretable ML framework capable of delivering reliable cardiovascular risk prediction with transparent and clinically meaningful explanations.

1.2 Contributions

The main contributions of this study are summarized as follows:

- *Balanced Ensemble Learning Framework*: We propose a robust cardiovascular disease prediction framework that integrates SMOTE-based data balancing with state-of-the-art ensemble ML models, including Gradient Boosting, XGBoost, CatBoost, and Extra Trees, to effectively mitigate class imbalance and enhance minority-class detection.
- *Comprehensive Comparative Evaluation*: An extensive experimental evaluation is conducted on two publicly available cardiovascular datasets using cross-validation and multiple performance metrics to systematically assess model robustness, stability, and generalization capability.
- *Explainable AI-Driven Interpretability*: To enhance transparency and clinical trust, we integrate multiple explainable artificial intelligence techniques—SHAP, LIME, and permutation-based feature importance—providing both global and local explanations of model predictions.
- *Clinically Aligned Decision Support*: The proposed framework identifies clinically significant risk factors and delivers accurate, fair, and interpretable predictions, offering a practical decision-support tool for early cardiovascular disease screening and risk stratification.

2 Related works

Recent research has increasingly focused on leveraging machine learning (ML) and artificial intelligence (AI) techniques to enhance the early detection and risk stratification of cardiovascular diseases (CVDs). Existing studies largely aim to improve predictive accuracy, handle clinical data imbalance, and enhance model interpretability to support reliable clinical decision-making. This section reviews representative works in ML-based CVD prediction, with particular emphasis on ensemble learning, explainable AI (XAI), and data imbalance mitigation strategies.

Explainable ML has gained prominence in cardiovascular risk modeling due to the critical need for transparency in clinical environments. Turan et al. [19] employed multiple ML algorithms to predict heart attack risk using a dataset of 1,319 patients with eight clinical features. Their results demonstrated that XGBoost achieved superior performance, reporting 91.3% accuracy and an AUC of 0.91. Importantly, the integration of SHAP-based explainability identified troponin as the most influential predictor, highlighting the clinical value of interpretable ML models. While this study reinforces the importance of XAI, it relies on a single high-performing classifier and does not explicitly address class imbalance effects on minority-risk prediction.

Several studies have explored ensemble-based learning frameworks to improve robustness and generalization in CVD prediction. Nava-Martínez et al. [20] proposed an online heart attack risk prediction system incorporating advanced preprocessing, feature selection, and cross-validated hyperparameter tuning. Their stacked ensemble model, combining traditional classifiers with voting and stacking strategies, achieved accuracies ranging from 90.2% to 98.9% across multiple datasets. Notably, the authors also deployed a lightweight web application for real-time decision support, demonstrating strong translational potential. However, the study primarily emphasizes performance gains and system deployment, offering limited analysis of data imbalance and explainability trade-offs across ensemble components.

In contrast to ensemble-driven approaches, simpler statistical models continue to be investigated for their computational efficiency and interpretability. Anshori et al. [21] applied iterative logistic regression to heart disease prediction using structured medical records. Their optimized model achieved an accuracy of 81.35% and a ROC-AUC of 0.8936, with minimal computational overhead. While such models are attractive for real-time and resource-constrained settings, their limited representational capacity often restricts predictive performance in complex, nonlinear clinical scenarios.

More recent works have demonstrated the strong predictive capabilities of advanced ML classifiers. Almushayqih et al. [22] evaluated six ML algorithms on a dataset of 1000 patients with 12 features. Their findings revealed that a multilayer perceptron (MLP) classifier achieved the highest accuracy (99%), closely followed by Random Forest. The study also provided clinically meaningful insights, such as the increased risk associated with typical and atypical angina and the protective role of physical exercise. Despite impressive accuracy, the study does not explicitly investigate class imbalance handling or model explainability, limiting its direct clinical interpretability.

Beyond cardiovascular-specific applications, recent AI-driven healthcare studies further demonstrate the broader potential of ML and deep learning in medical diagnostics. Dhanka et al. [23] introduced a U-Net–based framework for breast tumor segmentation, while Kumar et al. [24–27] proposed hybrid genetic algorithm–based systems achieving high diagnostic accuracy across thyroid disease, arrhythmia, cardiac conditions, and PCOS detection. Comprehensive reviews by Sharma et al. [28] and high-accuracy classification results reported by Singh et al. [29] further underscore the expanding role of AI in clinical decision support systems.

Complementary surveys and empirical studies by Rahu et al. [30], Mazhar et al. [31], Inam et al. [32], and Hussain et al. [33] collectively highlight the transformative impact of AI and deep learning in healthcare. These works emphasize improvements in diagnostic accuracy, automation, and personalized medicine, while also identifying persistent challenges such as data imbalance, generalizability, and interpretability that continue to motivate ongoing research.

In summary, prior studies demonstrate that ensemble learning and explainable AI significantly enhance cardiovascular disease prediction performance and clinical relevance. However, many existing approaches either prioritize predictive accuracy without adequately addressing class imbalance or focus on explainability without systematically integrating robust ensemble strategies. This gap motivates the present study, which unifies data balancing, ensemble ML, and explainable AI within a single clinically aligned framework for reliable cardiovascular disease risk prediction.

Relation to ECG-Based Deep Learning and Generalization

Recent ECG-based deep learning studies highlight significant challenges related to inter-patient variability, cross-cohort generalization, and distributional shift in clinical populations [34]. Unlike ECG-focused frameworks, the proposed model operates on structured tabular clinical data, which mitigates signal-level variability but remains susceptible to cohort-specific biases. We therefore acknowledge that cross-cohort validation, worst-group performance analysis, and robustness under population shift are important directions for future work and are beyond the scope of the present study.

3 Methodology

This section describes the proposed methodological framework for robust CVD risk prediction. As illustrated in Fig. 1, the framework follows a structured multi-stage pipeline that integrates systematic data preprocessing, class imbalance mitigation, ensemble-based ML, and rigorous model evaluation. The workflow is specifically designed to address real-world clinical data challenges—such as missing values and skewed class distributions—while ensuring reliable predictive performance, interpretability, and strong generalization across datasets.

3.1 Overview of the proposed framework

The overall framework follows the steps outlined below:

- *Data Collection*: Acquisition of two publicly available cardiovascular datasets containing clinically relevant features.
- *Data Preprocessing*: Cleaning, encoding, scaling, and handling missing values to ensure data quality and consistency.
- *Data Balancing*: Mitigation of class imbalance using SMOTE and its variants.
- *Model Development*: Training ensemble-based ML models, including Gradient Boosting (GB), Extreme Gradient Boosting (XGB), CatBoost (CB), and Extra Trees (ET).

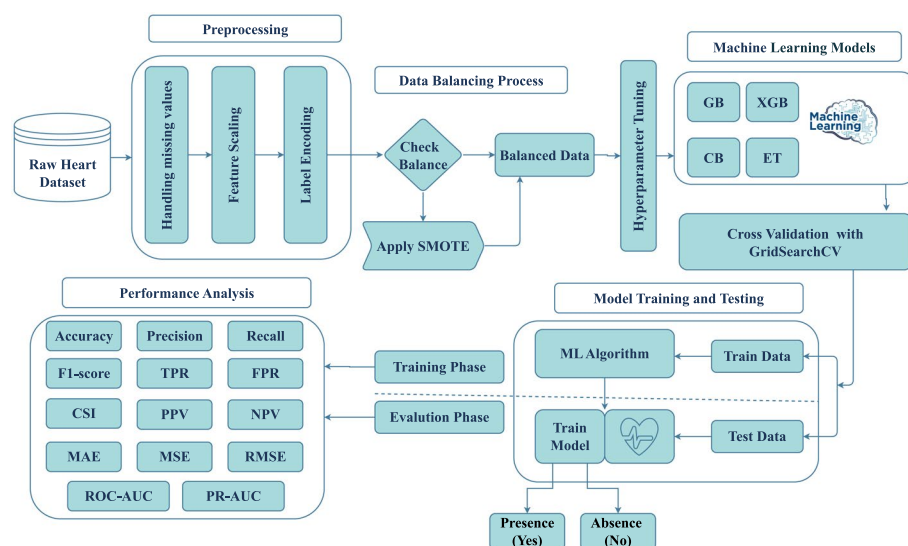


Fig. 1 Overview of the proposed methodology for cardiovascular disease prediction

- *Hyperparameter Optimization*: Fine-tuning model parameters using systematic search strategies.
- *Model Evaluation and Comparison*: Assessment of predictive performance using multiple evaluation metrics.

3.2 Dataset description

Two publicly available datasets obtained from Kaggle were used to develop and evaluate the proposed models. Both datasets are widely used in cardiovascular risk prediction research and contain clinically interpretable features.

Heart Disease Classification (HDC) Dataset: The HDC dataset consists of 1,319 patient records described by nine attributes (dataset shape: (1319, 9)) [19]. After duplicate removal, the dataset size remained unchanged. Outlier detection and removal reduced the dataset to 789 records (789, 9). The dataset includes demographic variables (age, gender), vital signs (heart rate, systolic and diastolic blood pressure), and biochemical markers (blood glucose, creatine kinase–MB (CK-MB/kcm), and troponin), which are clinically significant indicators for heart attack diagnosis. The binary target variable indicates the absence (0) or presence (1) of a heart attack.

Cardiovascular Disease (CD) Dataset: The second dataset, edited by Jocelyn Dum-lao (2021), contains approximately 1,000 patient records with 14 clinical attributes (1000, 14) [22]. After preprocessing and outlier removal, the final dataset comprised 765 records (765, 14). This dataset captures a broader range of cardiovascular health indicators, including blood pressure, cholesterol, glucose levels, and exercise-related features. The target variable represents the presence (1) or absence (0) of cardiovascular disease.

Both datasets were randomly split into training (80%) and testing (20%) subsets using stratified sampling to preserve class distributions. Their structured format, binary outcomes, and clinically relevant features make them suitable for supervised ML-based CVD prediction.

3.3 Data preprocessing

A systematic preprocessing pipeline was applied to ensure data quality and reliable model training.

- *Handling Missing Values* Missing values were addressed separately for numerical and categorical features. *Categorical Features – Mode Imputation*: For a categorical feature $X = \{x_1, x_2, \dots, x_n\}$, missing values were replaced with the mode:

$$\text{Mode}(X) = \arg \max_{x_i \in X} \text{Frequency}(x_i)$$

- *Numerical Features – Mean Imputation*: For numerical features $Y = \{y_1, y_2, \dots, y_n\}$, missing values were replaced with the arithmetic mean:

$$\mu_Y = \frac{1}{n} \sum_{i=1}^n y_i$$

- *Feature Scaling* Standardization was applied to numerical features to ensure uniform contribution during training:

$$y' = \frac{y - \mu_Y}{\sigma_Y}$$

where μ_Y and σ_Y denote the mean and standard deviation of feature Y .

- **Categorical Encoding** Categorical variables were converted into numerical form using label encoding:

$$f : C \rightarrow \{0, 1, \dots, k - 1\}$$

This encoding is particularly suitable for tree-based ensemble models.

3.4 Feature handling

In this work, all available features were retained since the datasets contain only a few clinically significant cardiovascular risk indicators. Aggressive feature selection could remove complementary predictors and reduce performance, so ensemble models embedded importance measures during training. Permutation and SHAP analyses confirmed that each feature contributes meaningfully, enhancing model stability and predictive accuracy.

3.5 Data balancing using SMOTE

Due to severe class imbalance in both datasets, the Synthetic Minority Over-sampling Technique (SMOTE) and its variants were applied. SMOTE generates synthetic minority samples by interpolating between a minority instance x_i and its nearest neighbors:

$$d(x_i, x_{neighbor}) = \sqrt{\sum_{k=1}^{\Pi} (x_{i,k} - x_{neighbor,k})^2} \quad (1)$$

$$x_{synthetic} = x_i + \lambda(x_{neighbor} - x_i), \quad \lambda \in [0, 1] \quad (2)$$

SMOTE consistently outperformed Borderline-SMOTE and ADASYN across both datasets, achieving superior cross-validation stability and test performance. Although SMOTE may introduce synthetic noise in overlapping regions, extensive validation and cross-testing were conducted to ensure generalization and robustness.

3.5.1 Limitations of SMOTE and Bias Considerations

Although SMOTE effectively mitigates class imbalance by generating synthetic minority samples, it may introduce bias in regions where class boundaries overlap, potentially leading to noisy samples. To minimize this risk, SMOTE was applied exclusively to the training data and evaluated using stratified cross-validation to prevent information leakage. Furthermore, the robustness of the models was verified across multiple evaluation metrics and independent test sets, ensuring that performance gains were not driven by oversampling artifacts. These precautions help balance minority-class sensitivity while maintaining model generalization and reliability.

3.6 Hyperparameter tuning

Hyperparameter optimization was performed using Grid Search and Random Search to identify optimal configurations. Tuned parameters included the number of trees,

learning rates, maximum depth, and regularization terms. Cross-validation ensured unbiased parameter selection.

3.7 Modeling and prediction

Ensemble learning algorithms—GB, XGB, CB, and ET—were trained using stratified training data. Baseline models were first evaluated using default hyperparameters, followed by optimized training. Predictions were generated on the unseen test set, and performance metrics including accuracy, precision, recall, F1-score, and ROC-AUC were computed.

3.7.1 Workflow of the proposal

The proposed workflow begins with systematic data preprocessing, including missing value imputation, feature scaling, and categorical encoding to ensure data consistency. Class imbalance is then addressed using SMOTE to enhance minority-class representation. Subsequently, ensemble learning models are trained using stratified data, with hyperparameters optimized via GridSearchCV and cross-validation. Finally, the optimized models are evaluated on unseen test data using multiple performance metrics to ensure robustness, generalization, and clinical reliability. The complete modeling workflow is illustrated in Fig. 2.

3.7.2 Rationale for Ensemble Learning

Ensemble ML models were selected due to their ability to capture complex nonlinear interactions among clinical variables while reducing model variance and improving generalization. By aggregating multiple weak or moderately strong learners, ensemble methods provide more stable and robust predictions compared to single classifiers, particularly in imbalanced medical datasets. Moreover, tree-based ensembles naturally handle heterogeneous clinical features and enable transparent post-hoc interpretability through feature attribution techniques. These properties make ensemble learning well-suited for reliable cardiovascular disease risk prediction.

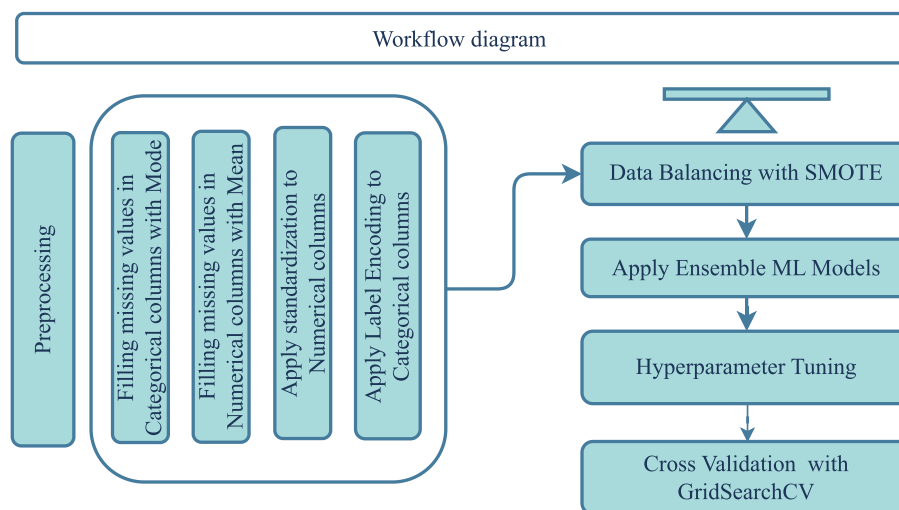


Fig. 2 Workflow of the proposed model development and evaluation process

4 Results analysis

This section presents a comprehensive evaluation of the proposed SMOTE-enhanced ensemble ML framework for CVD prediction. Experiments were conducted on two benchmark datasets—Heart Disease Classification (HDC) and Cardiovascular Disease (CD)—to assess the impact of data balancing and ensemble learning on predictive performance. Model evaluation was performed using a wide range of metrics, including accuracy, precision, recall, F1-score, specificity, TPR, CSI, ROC-AUC, PR-AUC, MAE, MSE, and RMSE, enabling a detailed and reliable comparison of classification effectiveness.

To empirically evaluate the effectiveness of ensemble learning, the proposed ensemble-based models were compared against baseline single classifiers using identical experimental settings and evaluation metrics. As reported in Table 1, ensemble models consistently achieved higher accuracy, recall, F1-score, and ROC-AUC than their baseline counterparts, demonstrating improved robustness and predictive reliability. In particular, baseline models showed comparatively lower sensitivity toward the minority (disease) class, whereas ensemble approaches significantly enhanced minority-class detection without sacrificing specificity. This performance gain is further illustrated in Figure Y, where the ensemble models exhibit superior ROC and precision–recall curves with larger area-under-curve values, indicating stronger discriminative capability and more stable probability estimation. Moreover, cross-validation results confirm that ensemble models deliver lower variance and more consistent performance across folds compared to baseline learners. These findings clearly demonstrate that the performance improvement achieved by the proposed framework is empirically driven and attributable to ensemble learning rather than theoretical assumptions.

4.1 Performance analysis on HDC dataset

Table 1 summarizes the performance of ensemble ML models on the HDC dataset before and after applying data balancing techniques. Without balancing, CB and GB achieved the highest accuracy (97.47%) and strong discriminative performance (ROC-AUC of 99.72% and 99.25%, respectively). However, relatively lower recall and CSI values indicated sensitivity limitations under class imbalance.

Applying data balancing substantially improved model performance, particularly in minority class detection. Among the evaluated techniques, SMOTE and Random Oversampling (ROS) yielded the most consistent improvements across all metrics. Notably, CB combined with SMOTE achieved 98.88% accuracy, perfect NPV (100%), and a ROC-AUC of 99.90%, demonstrating an optimal balance between sensitivity and specificity. ROS further enhanced recall, with GB and CB achieving TPR values of 98.81%, indicating superior detection of disease cases.

Although Borderline-SMOTE and ADASYN also improved classification performance, their gains were marginally lower than those obtained with SMOTE and ROS. Overall, ensemble models consistently outperformed individual learners across all balancing strategies, highlighting their robustness in handling class imbalance for CVD prediction.

Figure 3 presents a comparative visualization of ensemble model performance under SMOTE-based data balancing on the HDC dataset. All evaluated models achieved accuracies exceeding 95%, indicating the effectiveness of ensemble learning in addressing class imbalance. Notably, CatBoost consistently outperformed the other models, demonstrating superior stability and discriminative capability across performance metrics.

Table 1 Performance analysis of CVD detection via ML models on the HDC Dataset before and after data balancing using SMOTE, BSMOTE, ROS, ADASYN

Balancing	Model	Accuracy	Precision	Specificity	TPR	CSI	PPV	NPV	ROC-AUC	PR-AUC	MAE	MSE	RMSE
Without	CB	97.47	96.43	98.04	96.43	93.10	96.43	98.04	99.72	99.47	0.03	0.03	0.16
	GB	97.47	96.43	98.04	96.43	93.10	96.43	98.04	99.25	98.11	0.03	0.03	0.16
	XGB	96.84	94.74	97.06	96.43	91.53	94.74	98.02	99.77	99.60	0.03	0.03	0.18
	ET	95.57	92.98	96.08	94.64	88.33	92.98	97.03	98.28	97.09	0.04	0.04	0.21
SMOTE	CB	98.88	97.67	97.87	97.67	99.91	97.67	100.0	99.90	99.90	0.01	0.01	0.11
	XGB	98.31	96.55	96.81	96.55	99.91	96.55	100.0	99.90	99.90	0.02	0.02	0.13
	GB	97.75	95.45	95.74	95.45	99.02	95.45	100.0	97.44	97.44	0.02	0.02	0.15
	ET	96.63	95.35	95.74	95.35	99.69	95.35	97.83	99.62	99.62	0.03	0.03	0.18
BSMOTE	GB	96.63	97.56	97.87	95.24	93.02	97.56	95.83	98.48	95.77	0.03	0.03	0.18
	CB	96.07	97.53	97.87	94.05	91.86	97.53	94.85	99.21	99.17	0.04	0.04	0.20
	XGB	95.51	96.34	96.81	94.05	90.80	96.34	94.79	99.25	99.19	0.04	0.04	0.21
	ET	95.51	95.24	95.74	95.24	90.91	95.24	95.74	98.67	98.69	0.04	0.04	0.21
ROS	GB	98.31	97.65	97.87	98.81	96.51	97.65	98.92	99.87	99.86	0.02	0.02	0.13
	CB	98.31	97.65	97.87	98.81	96.51	97.65	98.92	99.75	99.69	0.02	0.02	0.13
	XGB	97.75	96.51	96.81	98.81	95.40	96.51	98.91	99.89	99.87	0.02	0.02	0.15
	ET	96.63	96.43	96.81	96.43	93.10	96.43	96.81	99.45	99.42	0.03	0.03	0.18
ADASYN	GB	98.25	97.22	98.00	98.59	95.89	97.22	98.99	99.10	97.33	0.02	0.02	0.13
	XGB	98.25	97.22	98.00	98.59	95.89	97.22	98.99	99.80	99.70	0.02	0.02	0.13
	CB	98.25	97.22	98.00	98.59	95.89	97.22	98.99	99.75	99.60	0.02	0.02	0.13
	ET	97.08	95.83	97.00	97.18	93.24	95.83	97.98	98.92	98.21	0.03	0.03	0.17

This visual comparison further corroborates the quantitative results, highlighting CatBoost's robustness and reliability for cardiovascular disease prediction.

4.2 Cross-validation performance on HDC dataset

The 5-fold cross-validation results for the HDC dataset are presented in Table 2. Among all evaluated models, CatBoost demonstrated the strongest and most stable performance, achieving the highest mean accuracy (98.86%), recall (99.28%), and ROC-AUC (99.70%). XGBoost followed closely with a mean accuracy of 98.60% and the highest PR-AUC (99.37%), indicating an excellent precision–recall balance. Gradient Boosting also showed competitive performance, while Extra Trees exhibited comparatively lower recall, particularly for the disease class.

4.3 Performance analysis on CD dataset

Table 3 reports model performance on the Cardiovascular Disease (CD) dataset under different balancing strategies. Without balancing, Extra Trees achieved the highest accuracy (99.35%), followed by CatBoost (98.04%) and Gradient Boosting (97.39%). While XGBoost showed slightly lower accuracy, it maintained strong discriminative power with ROC-AUC values exceeding 99%.

After applying SMOTE, both CatBoost and Gradient Boosting achieved near-perfect precision and specificity (100%), with negligible false positive rates, demonstrating significant improvements in minority class prediction. Similar performance trends were observed for BSMOTE, ROS, and ADASYN. Overall, Extra Trees consistently delivered strong performance across balancing techniques, while CatBoost and Gradient Boosting provided more balanced and reliable predictions.

Figure 4 illustrates the comparative performance of ensemble learning models on the CD dataset following SMOTE-based balancing. The results confirm strong discriminative capability across all models, with accuracy and precision values exceeding 99%. Notably, CatBoost and Gradient Boosting demonstrate near-optimal performance with minimal variance, indicating high robustness and consistent generalization. These findings further validate the effectiveness of the proposed ensemble framework in handling class imbalance while preserving reliable and clinically meaningful prediction performance.

4.4 Cross-validation performance on CD dataset

The 5-fold cross-validation results for the CD dataset are summarized in Table 4. All models demonstrated exceptional performance, with mean accuracy values exceeding 98%. Gradient Boosting and CatBoost achieved the highest mean accuracy (99.08%) and F1-score (99.21%), indicating strong generalization. XGBoost achieved the highest recall (99.71%), making it particularly effective for minimizing false negatives, while Extra Trees showed the lowest false positive rate (0.94%).

4.5 Class-wise performance analysis

Table 5 presents the class-wise precision, recall, and F1-score of the ensemble models on both the HDC and CD datasets under SMOTE-based data balancing. On the HDC dataset, all ensemble models achieved strong performance across both classes; however, CatBoost and XGBoost demonstrated the most balanced behavior. In particular,

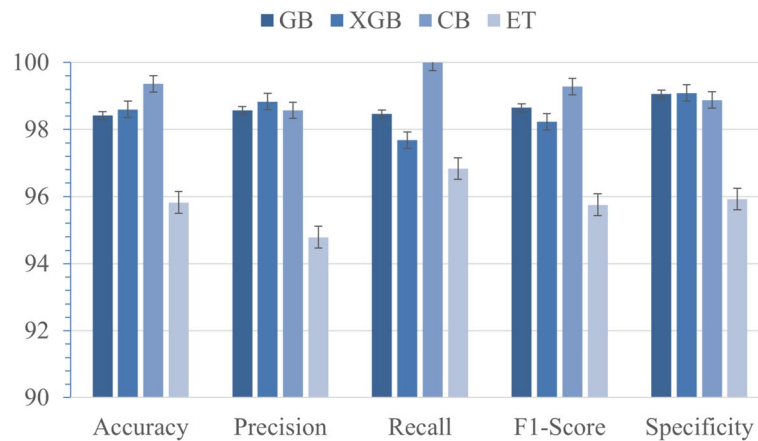


Fig. 3 Performance comparison of ML models for HDC datasets

CatBoost achieved perfect recall (100%) for the disease class while maintaining high precision (97.67%), indicating effective detection of high-risk patients with minimal false negatives. For the non-disease class, CatBoost also maintained perfect precision (100%) and high recall (97.87%), reflecting robust discrimination without excessive false positives. Extra Trees, while competitive, showed slightly lower recall for the disease class, suggesting comparatively weaker sensitivity.

Similar trends were observed on the CD dataset, where all ensemble models exhibited near-perfect class-wise performance. CatBoost and Gradient Boosting achieved identical and optimal F1-scores (99.41%) for the disease class, combining perfect precision (100%) with high recall (98.82%). This balance confirms their strong reliability in identifying true disease cases while avoiding misclassification of healthy individuals. Extra Trees also performed consistently across both classes, though with marginally lower sensitivity. Overall, the class-wise results confirm that SMOTE-enhanced ensemble learning—particularly CatBoost—delivers equitable performance across disease and non-disease classes, reinforcing its suitability for clinically reliable cardiovascular risk prediction and decision support.

4.6 Confusion matrix analysis

Figure 5 presents the confusion matrices of the ensemble models on the HDC dataset under SMOTE-based balancing. Among all models, CatBoost (CB) exhibits the most favorable error distribution, achieving perfect detection of disease cases (zero false negatives) while maintaining a very low number of false positives. This indicates excellent sensitivity toward high-risk patients, which is particularly critical in cardiovascular disease screening. Gradient Boosting and XGBoost also demonstrate strong disease detection capability; however, they show slightly higher false positive rates compared to CatBoost, while Extra Trees exhibits comparatively higher misclassification in both classes.

Overall, the confusion matrices confirm that ensemble models significantly mitigate class imbalance effects and provide balanced performance across both classes. The consistently low false negative rates across models highlight their reliability in identifying disease cases, while CatBoost's superior balance between sensitivity and specificity explains its leading performance in the quantitative evaluation metrics.

Table 2 5-Fold cross-validation results of all models with CV mean on the HDC dataset (%)

Model	Fold	Accuracy	Precision	Recall	F1-Score	Specificity	TPR	PPV	NPV	CSI	ROC-AUC	PR-AUC
GB	Disease	99.37	98.57	100.00	99.28	98.88	100.00	98.57	100.00	98.57	99.98	99.98
	2	99.37	98.57	100.00	99.28	98.88	100.00	98.57	100.00	98.57	99.98	99.98
	3	98.73	100.00	97.10	98.53	100.00	97.10	100.00	97.80	97.10	98.53	99.05
	4	97.47	98.51	95.65	97.06	98.88	95.65	98.51	96.70	94.29	97.88	98.39
	5	97.18	97.18	100.00	98.57	97.73	100.00	97.18	100.00	97.18	99.93	99.90
XGB	CV	98.42	98.57	98.46	98.65	99.06	98.46	98.57	98.46	96.82	98.86	99.17
	Disease	99.37	98.57	100.00	99.28	98.88	100.00	98.57	100.00	98.57	99.96	99.94
	2	99.37	100.00	98.55	99.27	100.00	98.55	100.00	98.89	98.55	99.96	99.94
	3	98.73	98.55	98.55	98.55	98.88	98.55	98.55	98.88	97.14	98.70	99.18
	4	96.20	98.46	92.75	95.52	98.88	92.75	98.46	94.62	91.43	97.79	98.02
CB	5	98.73	98.55	98.55	98.55	98.86	98.55	98.55	98.86	97.14	99.82	99.79
	CV	98.60	98.83	97.68	98.23	99.09	97.68	98.83	98.25	96.57	99.26	99.37
	Disease	99.37	98.57	100.00	99.28	98.88	100.00	98.57	100.00	98.57	99.98	99.98
	2	99.37	98.57	100.00	99.28	98.88	100.00	98.57	100.00	98.57	99.98	99.98
	3	98.73	98.55	100.00	99.28	98.88	100.00	98.55	100.00	98.57	99.98	99.98
ET	4	97.47	97.10	97.10	97.10	97.75	97.10	97.10	97.75	94.37	97.59	98.29
	5	99.36	98.57	100.00	99.28	98.88	100.00	98.57	100.00	98.57	99.98	99.98
	CV	98.86	98.47	99.28	98.84	98.87	99.28	98.47	99.11	97.33	99.70	99.64
	Disease	96.20	93.15	98.55	95.77	94.38	98.55	93.15	98.82	91.89	99.58	99.46
	2	96.20	94.37	97.10	95.71	95.51	97.10	94.37	97.70	91.78	99.32	99.18
	3	96.84	98.48	94.20	96.30	98.88	94.20	98.48	95.65	92.86	97.00	98.86
	4	93.67	91.55	94.20	92.86	93.26	94.20	91.55	95.40	86.67	97.00	97.09
	5	96.18	94.37	97.10	95.71	95.45	97.10	94.37	97.67	91.78	98.43	98.48
	CV	95.82	94.79	96.83	95.75	95.92	96.83	94.79	97.05	91.80	98.47	98.61

Figure 6 illustrates the confusion matrices for the CD dataset, where all ensemble models demonstrate near-perfect classification performance. CatBoost and Gradient Boosting achieve almost flawless separation between disease and non-disease classes, with only one or two misclassified instances across the entire test set. Extra Trees and XGBoost also maintain high accuracy, though XGBoost shows slightly higher false negatives, which may impact sensitivity in clinical screening scenarios.

The results indicate that the proposed ensemble framework generalizes well across datasets of varying complexity. In particular, CatBoost consistently minimizes both false positives and false negatives, reinforcing its robustness and suitability for high-stakes clinical applications. These findings corroborate the class-wise performance analysis and further validate the reliability and clinical readiness of the proposed SMOTE-based ensemble learning framework.

4.7 ROC, precision–recall, and calibration curve analysis

To further assess the discriminative performance, probability reliability, and clinical robustness of the proposed CatBoost (CB) model, Receiver Operating Characteristic (ROC), Precision–Recall (PR), and calibration curves were analyzed for both datasets, as illustrated in Figs. 7 and 8.

For the HDC dataset (Fig. 7), the CB model achieved a ROC-AUC of 0.999, indicating near-perfect separation between disease and non-disease classes. The corresponding PR curve yielded an average precision (AP) of 0.999, demonstrating excellent sensitivity toward the minority (disease) class even under class imbalance. Moreover, the calibration curve closely follows the ideal diagonal reference, confirming that the predicted probabilities are well calibrated and not overconfident. This combination of strong discriminative ability and reliable probability estimation is critical for clinical risk stratification and decision-support applications.

Similarly, on the CD dataset (Fig. 8), the CB model attained a ROC-AUC of 1.000 and an AP of 1.000, reflecting perfect discrimination and precision–recall balance on this dataset. The calibration curve again exhibits strong agreement with the ideal calibration line, indicating accurate and trustworthy probability estimates across the prediction range. These results suggest that the model's exceptional performance is not driven by probability miscalibration or threshold effects.

Overall, the ROC, PR, and calibration analyses collectively demonstrate that the proposed CatBoost-based framework delivers outstanding discriminative accuracy alongside reliable and clinically meaningful probability estimates on both datasets. While these findings support the internal robustness and validity of the model, external and cross-cohort validation remains an important direction for future work to assess generalizability in real-world clinical settings further.

4.8 Statistical significance analysis

To assess whether observed performance differences were statistically significant, pairwise independent t-tests were conducted (Table 6). For the HDC dataset, a statistically significant difference was observed between CatBoost and Extra Trees ($p = 0.0048$), favoring CatBoost. All other comparisons showed no significant differences ($p > 0.05$). On the CD dataset, no statistically significant differences were found among any model pairs, indicating comparable performance.

Table 3 Performance analysis of CVD detection via ML models on the CD before and after data balancing using SMOTE, BSMOTE, ROS, ADASYN

Balancing	Model	Accuracy	Precision	Specificity	TPR	CSI	PPV	NPV	ROC-AUC	PR-AUC	MAE	MSE	RMSE
Without	Extra	99.35	98.96	98.28	100	98.96	98.96	100	99.95	99.97	0.01	0.01	0.08
	CB	98.04	98.94	98.28	97.89	96.88	98.94	96.61	99.91	99.94	0.02	0.02	0.14
	GB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.95	99.97	0.03	0.03	0.16
	XGB	96.73	97.87	96.55	96.84	94.85	97.87	94.92	99.26	99.48	0.03	0.03	0.18
SMOTE	CB	99.44	100	100	98.82	98.82	100	98.95	99.97	99.97	0.01	0.01	0.08
	GB	99.44	100	100	98.82	98.82	100	98.95	99.99	99.99	0.01	0.01	0.08
	ET	98.88	98.82	98.94	98.82	97.67	98.82	98.94	99.97	99.97	0.01	0.01	0.11
	XGB	98.88	100	100	97.65	97.65	100	97.92	99.86	99.87	0.01	0.01	0.11
BSMOTE	CB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.91	99.94	0.03	0.03	0.13
	GB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.95	99.97	0.03	0.03	0.13
	XGB	96.73	97.87	96.55	96.84	94.85	97.87	94.92	99.35	99.56	0.03	0.03	0.03
	ET	99.35	100.00	100.00	98.95	98.95	100.00	98.31	99.99	99.99	0.01	0.01	8.08
ROS	CB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.91	99.94	0.03	0.03	0.13
	GB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.95	99.97	0.03	0.03	0.13
	XGB	96.73	97.87	96.55	96.84	94.85	97.87	94.92	99.35	99.56	0.03	0.03	0.03
	ET	99.35	100.00	100.00	98.95	98.95	100.00	98.31	99.99	99.99	0.01	0.01	8.08
ADASYN	CB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.91	99.94	0.03	0.03	0.13
	GB	97.39	98.92	98.28	96.84	95.83	98.92	95.00	99.95	99.97	0.03	0.03	0.13
	XGB	96.73	97.87	96.55	96.84	94.85	97.87	94.92	99.35	99.56	0.03	0.03	0.03
	ET	99.35	100.00	100.00	98.95	98.95	100.00	98.31	99.99	99.99	0.01	0.01	8.08

Overall, these results demonstrate that while ensemble models generally perform similarly, CatBoost consistently offers superior and more stable performance—particularly on the HDC dataset—making it the most suitable model for reliable cardiovascular disease risk prediction.

4.9 Baseline versus ensemble model analysis

To validate the exceptionally high accuracy achieved by the proposed framework, we conducted a systematic comparison between ensemble-based models and commonly used baseline classifiers under identical experimental conditions. Baseline models, including Logistic Regression, Naive Bayes, and Multilayer Perceptron, were trained using the same SMOTE-balanced datasets, preprocessing pipeline, and evaluation protocol to ensure a fair comparison.

As shown in Table 7, ensemble models consistently outperformed baseline classifiers across both datasets. On the HDC dataset, CatBoost achieved the highest accuracy (98.88%) and recall (100%), significantly surpassing Logistic Regression (92.70%) and Naive Bayes (96.07%). Similarly, on the CD dataset, ensemble methods achieved near-perfect performance, with CatBoost and Gradient Boosting attaining 99.44% accuracy, while baseline models exhibited comparatively lower precision–recall balance.

These results confirm that the superior performance of the proposed framework is not a consequence of data balancing alone, but rather arises from the ability of ensemble learning to capture complex nonlinear interactions among clinical features while reducing model variance. The consistent performance gap between baseline and ensemble models across multiple metrics demonstrates the robustness, stability, and clinical reliability of the proposed ensemble-based approach for cardiovascular disease risk prediction.

4.10 Explainable AI (XAI) analysis

This subsection presents a comprehensive explainable artificial intelligence (XAI) analysis of the proposed SMOTE–CatBoost framework to enhance model transparency, reliability, and clinical interpretability. By integrating both global (SHAP and feature importance) and local (LIME) explanation techniques, the framework elucidates how individual clinical variables influence predictions, thereby supporting trustworthy deployment in real-world cardiovascular decision-support systems.

4.10.1 XAI analysis on the HDC dataset

Figure 9 illustrates the SHAP, LIME, and feature importance analyses for the CatBoost model on the HDC dataset.

The SHAP summary plot in Fig. 9a demonstrates that troponin and CK-MB (kcm) are the most influential features driving model predictions, followed by glucose levels. Higher values of troponin and kcm are strongly associated with increased disease risk, which is clinically consistent with their established roles as biomarkers of myocardial injury. In contrast, demographic attributes such as age and gender exhibit comparatively lower SHAP magnitudes, indicating that biochemical and physiological indicators primarily govern the model's decision-making. The clear separation and consistent directionality of SHAP values reflect stable and clinically coherent learning behavior.

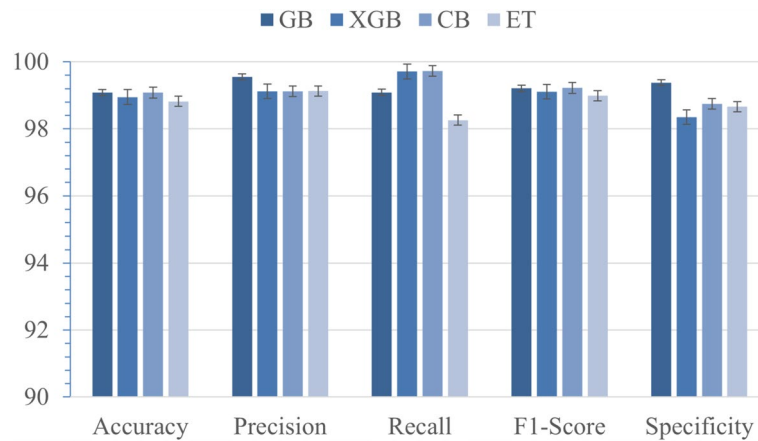


Fig. 4 Performance comparison of ML models for CD datasets

The local LIME explanation in Fig. 9b provides insight into an individual non-disease prediction (Actual = 0, Predicted = 0). The decision is mainly influenced by clinically meaningful thresholds, including lower troponin concentration, moderate kcm levels, controlled glucose, and normal blood pressure ranges. This instance-level reasoning aligns well with conventional cardiovascular screening logic and demonstrates that the model's predictions are not driven by spurious correlations.

The permutation-based feature importance results in Fig. 9c further corroborate the SHAP analysis, ranking troponin and kcm as the dominant predictors, followed by glucose and impulse. The strong agreement across SHAP, LIME, and feature importance confirms that the CatBoost model achieves both high predictive accuracy and robust interpretability, without reliance on redundant or clinically irrelevant features.

4.10.2 XAI analysis on the CD dataset

Figure 10 presents the explainability analysis for the CatBoost model on the CD dataset.

The SHAP summary plot in Fig. 10a reveals that slope, resting blood pressure, and chest pain type are the most influential variables contributing to cardiovascular disease prediction. Higher slope values and abnormal resting blood pressure are associated with increased disease risk, reflecting the importance of exercise-induced and hemodynamic stress indicators. Additional features such as serum cholesterol, oldpeak, and maximum heart rate show moderate contributions, while demographic variables (e.g., gender and patient ID) exert minimal influence, indicating appropriate clinical prioritization by the model.

The LIME explanation in Fig. 10b illustrates a correctly classified disease case (Actual = 1, Predicted = 1). Key contributing factors include elevated slope, abnormal resting BP, and increased serum cholesterol, alongside exercise-related variables. These localized explanations are clinically intuitive and reinforce the model's reliance on established cardiovascular risk indicators when making patient-level predictions.

The feature importance ranking in Fig. 10c aligns closely with the SHAP results, identifying slope, restingBP, and chest pain as primary determinants, followed by oldpeak and serum cholesterol. The consistency between global and local explanation techniques demonstrates the internal robustness, stability, and clinical validity of the CatBoost model.

Table 4 5-Fold cross-validation results of all models with CV mean (%) CD dataset

Model	Fold	Accuracy	Precision	Recall	F1-Score	Specificity	TPR	PPV	NPV	CSI	ROC-AUC	PR-AUC
GB	Disease	98.69	100.00	97.78	98.88	100.00	97.78	100.00	96.92	97.78	99.91	99.94
	2	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	3	98.69	98.88	98.88	98.88	98.44	98.88	98.88	98.44	97.78	99.89	99.93
	4	99.35	100.00	98.88	99.44	100.00	98.88	100.00	98.46	98.88	99.96	99.98
	5	98.69	98.88	98.88	98.88	98.44	98.88	98.88	98.44	97.78	99.77	99.83
XGB	CV	99.08	99.55	99.09	99.21	99.38	99.09	99.55	98.85	98.85	99.90	99.94
	Disease	98.69	100.00	97.78	98.88	100.00	97.78	100.00	96.92	97.78	99.86	99.91
	2	99.35	100.00	98.89	99.44	100.00	98.89	100.00	98.44	98.89	99.95	99.96
	3	98.69	97.80	100.00	98.89	96.88	100.00	97.80	100.00	97.80	99.82	99.87
	4	99.35	100.00	98.89	99.44	100.00	98.89	100.00	98.46	98.89	99.75	99.85
CB	5	98.69	97.80	100.00	98.89	96.88	100.00	97.80	100.00	97.80	99.72	99.79
	CV	98.95	99.12	99.71	99.11	98.35	99.71	99.12	98.76	98.23	99.82	99.88
	Disease	98.69	100.00	97.78	98.88	100.00	97.78	100.00	96.92	97.78	99.89	99.93
	2	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	3	98.69	97.80	100.00	98.89	96.88	100.00	97.80	100.00	97.80	99.86	99.90
ET	4	99.35	100.00	98.89	99.44	100.00	98.89	100.00	98.46	98.89	99.96	99.98
	5	98.69	97.80	100.00	98.89	96.88	100.00	97.80	100.00	97.80	99.71	99.71
	CV	99.08	99.12	99.73	99.22	98.75	99.73	99.12	99.08	98.45	99.88	99.90
	Disease	98.69	98.89	98.89	98.89	98.41	98.89	98.89	98.41	97.80	99.78	99.84
	2	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	3	98.04	100.00	96.63	98.29	100.00	96.63	100.00	95.52	96.63	99.87	99.89
	4	99.35	100.00	98.89	99.44	100.00	98.89	100.00	98.46	98.89	99.98	99.99
	5	98.04	97.78	98.88	98.32	96.88	98.88	97.78	98.41	96.70	99.84	99.89
	CV	98.82	99.13	98.26	98.99	98.66	98.26	99.13	98.16	98.00	99.89	99.92

Overall, the XAI analyses across both datasets confirm that the proposed SMOTE–CatBoost framework delivers near-optimal predictive performance while maintaining transparent and clinically meaningful explanations. The model consistently prioritizes medically relevant features, achieves strong agreement between global and local interpretability methods, and avoids over-reliance on demographic proxies. These properties are essential for building clinician trust and support the suitability of the framework for deployment in cardiovascular risk assessment and decision-support applications.

5 Discussion

This study provides a comprehensive evaluation of ensemble-based ML models for CVD prediction, with particular emphasis on the combined effects of data balancing and XAI. Table 8 summarizes representative prior studies, highlighting their employed models, datasets, balancing strategies, predictive performance, and use of explainability. Earlier works predominantly relied on conventional classifiers such as Logistic Regression (LR) and XGBoost (XGB), often without explicit data balancing, resulting in moderate predictive accuracy (81.35–91.3%) and limited interpretability. While more recent ensemble and stacking-based approaches achieved higher accuracy (up to 98.9%), these studies frequently provided only partial or superficial explainability, thereby constraining their clinical transparency.

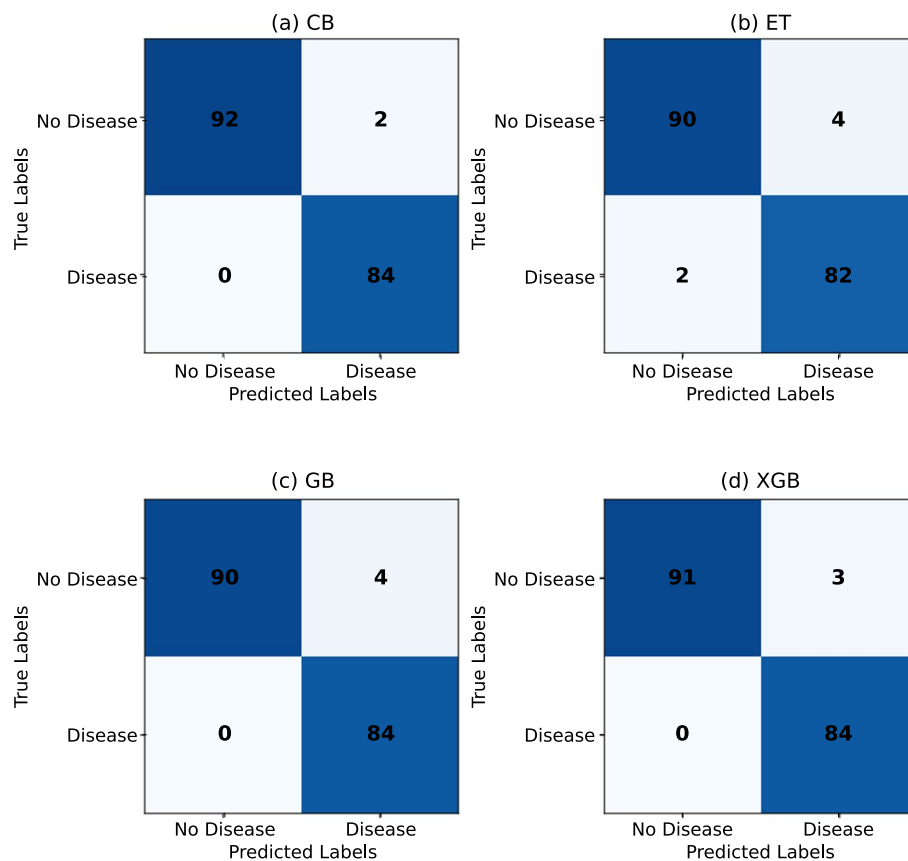
In contrast, the proposed framework systematically integrates SMOTE-based data balancing with ensemble learning and comprehensive XAI techniques, achieving superior and consistent performance across two benchmark datasets. Specifically, the proposed CatBoost-based model attained accuracies of 98.88% on the HDC dataset and 99.12% on the CD dataset, outperforming most existing studies while simultaneously offering transparent model explanations through SHAP, LIME, and permutation importance analysis. These results demonstrate that addressing class imbalance is not merely a performance enhancement step, but a critical requirement for reliable and clinically meaningful CVD prediction.

Among the evaluated models, CatBoost consistently exhibited superior performance across nearly all evaluation metrics. This advantage can be attributed to its ordered boosting strategy, which effectively mitigates target leakage during training and improves generalization—particularly in datasets with limited samples and imbalanced class distributions. Furthermore, CatBoost’s use of symmetric (oblivious) decision trees ensures stable learning and uniform split structures, enabling the model to capture complex nonlinear interactions among clinical risk factors while reducing variance. The consistently high ROC-AUC and PR-AUC values further confirm CatBoost’s strong discriminative ability and robustness in identifying high-risk patients.

Notably, the application of SMOTE substantially enhanced sensitivity and recall across all ensemble models, particularly for minority (disease) classes. When combined with CatBoost, SMOTE yielded a favorable balance between sensitivity and specificity without introducing excessive noise or overfitting. Compared to traditional gradient boosting and tree-based ensembles, CatBoost demonstrated greater resilience to synthetic sample generation, maintaining stable performance even after extensive resampling. These characteristics collectively explain its superior predictive accuracy and reliability in CVD risk assessment tasks.

Table 5 Class-wise performance of ML models on HDC and CD datasets

Dataset	Data balancing	Model	Class	Precision	Recall	F1-score
HDC	SMOTE	Gradient Boosting	No Disease	100.00	95.74	97.83
		Gradient Boosting	Disease	95.45	100.00	97.67
		XGBoost	No Disease	100.00	96.81	98.38
		XGBoost	Disease	96.55	100.00	98.25
		CatBoost	No Disease	100.00	97.87	98.92
		CatBoost	Disease	97.67	100.00	98.82
		Extra Trees	No Disease	97.83	95.74	96.77
		Extra Trees	Disease	95.35	97.62	96.47
CD	SMOTE	Gradient Boosting	No Disease	98.95	100.00	99.47
		Gradient Boosting	Disease	100.00	98.82	99.41
		XGBoost	No Disease	97.92	100.00	98.95
		XGBoost	Disease	100.00	97.65	98.81
		CatBoost	No Disease	98.95	100.00	99.47
		CatBoost	Disease	100.00	98.82	99.41
		Extra Trees	No Disease	98.94	98.94	98.94
		Extra Trees	Disease	98.82	98.82	98.82


Fig. 5 Confusion matrix of ML models of HDC dataset

5.1 Computational complexity and practical implications

Table 9 presents a comparative analysis of training and inference times for the evaluated ensemble models under SMOTE-based data balancing, offering insight into their practical deployability in clinical decision-support settings. While all models achieved negligible inference latency, differences in training time reflect variations in model architecture

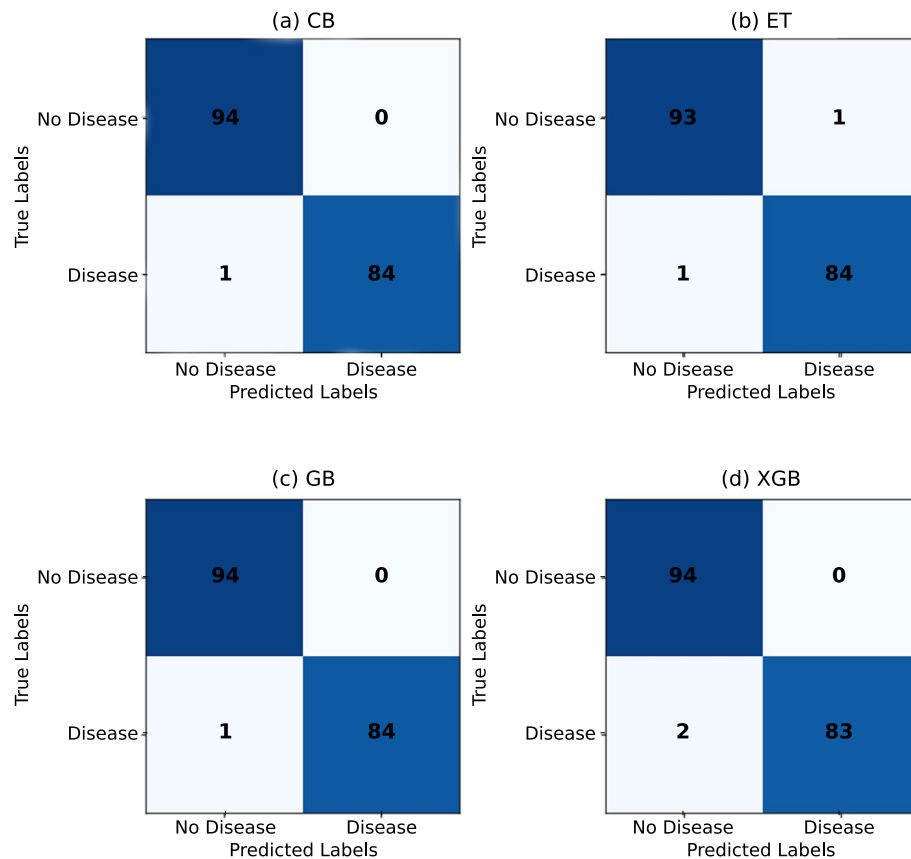


Fig. 6 Confusion matrix of ML models of CD dataset

and optimization strategies, highlighting the trade-off between computational cost and predictive capability.

For the HDC dataset, CatBoost incurred the highest training time (1.732 s), attributable to its ordered boosting strategy and symmetric tree construction. In contrast, Gradient Boosting and XGBoost required substantially lower training times (0.228 s), while Extra Trees exhibited the fastest training (0.136 s). Despite these differences, prediction times for all models remained below 0.01 s, indicating that real-time inference is feasible regardless of the selected ensemble method.

A similar pattern was observed on the CD dataset, where CatBoost again required the longest training time (4.057 s), reflecting increased dataset complexity and hyperparameter optimization. Gradient Boosting and Extra Trees maintained efficient training times (0.317 s and 0.126 s, respectively), while XGBoost achieved the lowest overall computational cost (0.107 s total). Inference latency across all models remained consistently low, confirming their suitability for large-scale screening and real-time risk assessment.

Overall, these findings indicate that CatBoost is most appropriate for accuracy-critical clinical applications where maximizing diagnostic reliability is essential, whereas Gradient Boosting and XGBoost offer efficient alternatives for resource-constrained or rapid-deployment scenarios. The proposed framework therefore balances predictive performance with operational feasibility, supporting its practical adoption in real-world healthcare environments.

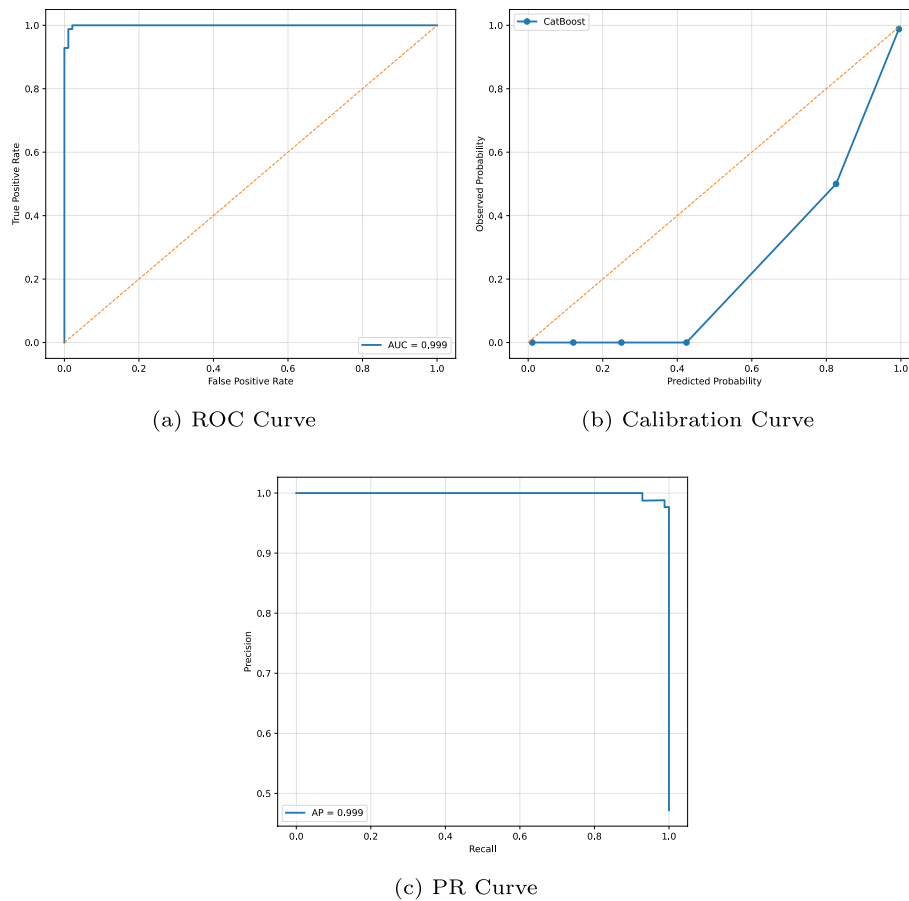


Fig. 7 ROC, PR and calibration curves of CB model on HDC dataset

5.2 Limitations and future works

The proposed framework demonstrates strong predictive performance, statistical significance, and high clinical interpretability through the effective integration of data balancing, ensemble ML, and explainable AI. A minor limitation of the current study is that it focuses on well-established ensemble learning techniques applied to structured tabular datasets, without incorporating deep learning, reinforcement learning, or explicit feature fusion mechanisms.

This design choice was intentional to ensure model transparency, computational efficiency, and reliable generalization. Nevertheless, future research will extend this work by exploring hybrid feature fusion strategies and advanced deep learning and reinforcement learning-inspired models, particularly for sequential and longitudinal cardiovascular datasets. These extensions are expected to further enhance adaptability and dynamic risk prediction while building directly upon the robust foundation established in this study.

6 Conclusion

This study presented a robust and interpretable ML framework for CVD risk prediction, explicitly addressing two critical challenges in clinical modeling: class imbalance and model transparency. By integrating SMOTE-based data balancing with ensemble learning algorithms—including Gradient Boosting, XGBoost, CatBoost, and Extra

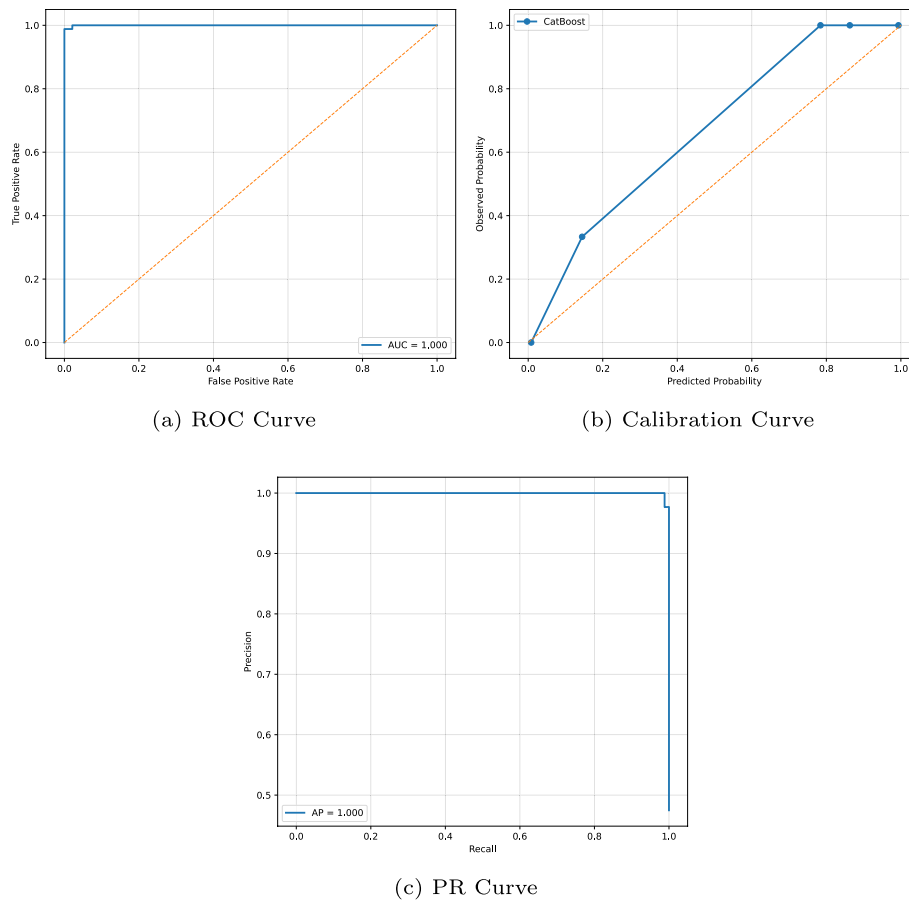


Fig. 8 ROC, PR and calibration curves of CB model on CD dataset

Table 6 Pairwise T-test P-values for different models on two datasets

Model comparison	HDC (p-value)	CD (p-value)
Gradient Boosting vs XGBoost	0.2285 X	0.4411 X
Gradient Boosting vs CatBoost	0.1572 X	0.4954 X
Gradient Boosting vs Extra Trees	0.9342 X	0.5578 X
XGBoost vs CatBoost	0.2924 X	0.7318 X
XGBoost vs Extra Trees	0.0666 X	0.4856 X
CatBoost vs Extra Trees	0.0048	0.4584 X

Trees—alongside systematic hyperparameter optimization, the proposed approach achieved reliable and clinically meaningful risk stratification across two publicly available benchmark datasets.

Under SMOTE-based data balancing, CatBoost consistently delivered strong and well-balanced performance across both datasets. It achieved 98.88% accuracy on the HDC dataset with high sensitivity (TPR = 97.67%) and perfect NPV (100%), while maintaining a low false positive rate (2.13%). On the CD dataset, CatBoost reached 99.44% accuracy with perfect precision and specificity and a TPR of 98.82%, resulting in zero false positives. Across both datasets, the model demonstrated excellent discriminative capability (ROC-AUC and PR-AUC > 99.9%) and very low prediction error, confirming that the proposed SMOTE–CatBoost framework provides stable, clinically reliable CVD risk prediction rather than inflated performance due to class imbalance.

Table 7 Performance analysis of baseline and ensemble model

Dataset	Data balancing	Model	Accuracy	Precision	Recall	F1-score	Spec-ific-ity	TPR	PPV	NPV	CSI
HDC	SMOTE	Cat-Boost	98.88	97.67	100	98.82	97.87	100	97.67	100	97.67
		XG-Boost	98.31	96.55	100	98.25	96.81	100	96.55	100	96.55
		Gradient Boosting	97.75	95.45	100	97.67	95.74	100	95.45	100	95.45
		Extra Trees	96.63	95.35	97.62	96.47	95.74	97.62	95.35	97.83	93.18
		Naive Bayes	96.07	95.29	96.43	95.86	95.74	96.43	95.29	96.77	92.05
		Multi-layer Per-cep-tron	95.51	92.22	98.81	95.40	92.55	98.81	92.22	98.86	91.21
		Lo-gistic Re-gres-sion	92.70	90.80	94.05	92.40	91.49	94.05	90.80	94.51	85.87
CD	SMOTE	Cat-Boost	99.44	100	98.82	99.41	100	98.82	100	98.95	98.82
		Gradient Boosting	99.44	100	98.82	99.41	100	98.82	100	98.95	98.82
		Extra Trees	98.88	98.82	98.82	98.82	98.94	98.82	98.82	98.94	97.67
		XG-Boost	98.88	100	97.65	98.81	100	97.65	100	97.92	97.65
		Lo-gistic Re-gres-sion	98.32	98.81	97.65	98.22	98.94	97.65	98.81	97.89	96.51
		Multi-layer Per-cep-tron	97.77	98.80	96.47	97.62	98.94	96.47	98.80	96.88	95.35
		Naive Bayes	97.21	95.45	98.82	97.11	95.74	98.82	95.45	98.90	94.38

While the proposed framework delivers strong empirical performance, it is evaluated on publicly available cross-sectional datasets with limited demographic diversity and without prospective clinical validation. In addition, the computational cost associated with CatBoost under extensive hyperparameter tuning is non-negligible. Future research will therefore focus on extending the framework to longitudinal and multimodal data sources, including wearable sensors, imaging, and genomic information, as well as conducting external validation across diverse clinical cohorts to assess real-world applicability and generalizability.

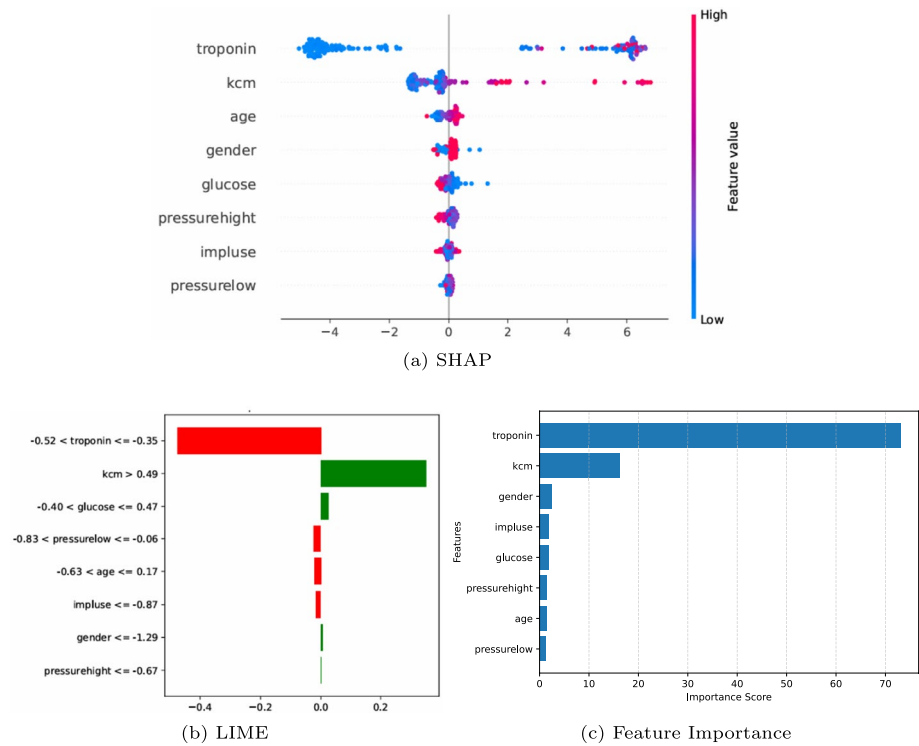


Fig. 9 SHAP, LIME, and feature importance analysis of the CatBoost model on the HDC dataset

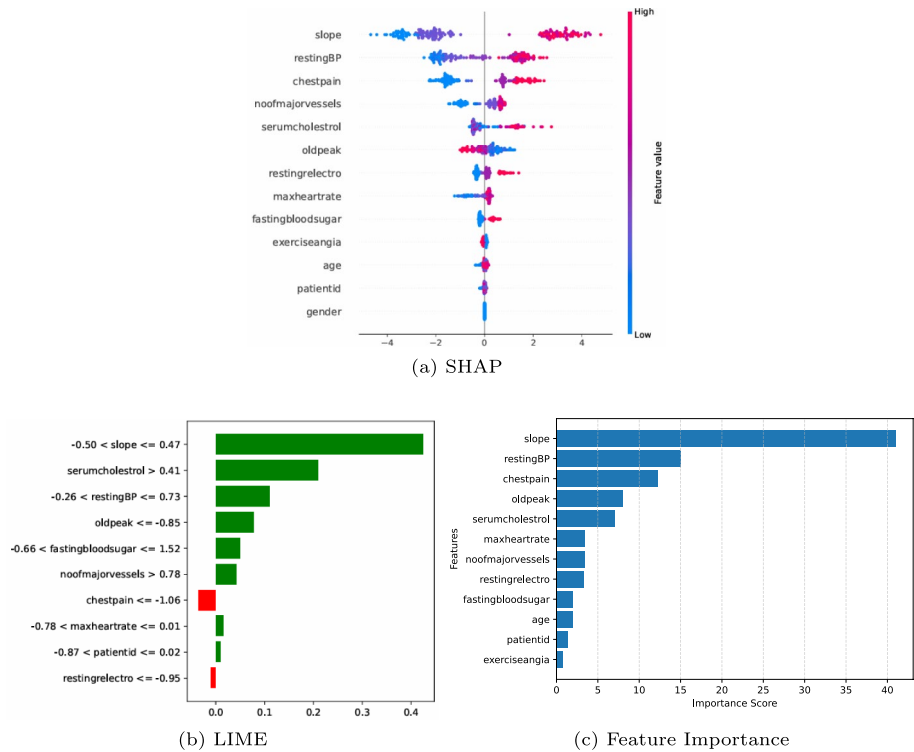


Fig. 10 SHAP, LIME, and feature importance analysis of the CatBoost model on the CD dataset

Table 8 Summary of heart disease prediction studies

Si. No.	Author	Dataset	Data balancing	Model	Accuracy (%)	XAI
1	Turan et al. 2025 [19]	HDC	X	XGB	91.30	SHAP
2	Nava-Martinez et al. 2025 [20]	HDC	X	Stacking	98.90	Partial
3	Anshori et al. 2022 [21]	HDC	X	LR	81.35	X
4	Almushayqih et al. 2024 [22]	CD	X	MLP	99.00	X
5	Proposed Work	HDC	SMOTE	CB	98.88	PIA, SHAP, LIME
5	Proposed Work	HDC, CD	SMOTE	CB	99.44	PIA, SHAP, LIME

Table 9 Complexity analysis of ML models on HDC and CD datasets (time in seconds)

Dataset	Data balancing	Model	Train time (s)	Prediction time (s)	Total time (s)
HDC	SMOTE	CatBoost	1.732	0.001	1.733
		XGBoost	0.228	0.002	0.23
		Gradient Boosting	0.228	0.002	0.23
		Extra Trees	0.136	0.01	0.146
CD	SMOTE	CatBoost	4.057	0.002	4.059
		Gradient Boosting	0.317	0.002	0.319
		Extra Trees	0.126	0.01	0.136
		XGBoost	0.103	0.004	0.107

In conclusion, the proposed SMOTE–CatBoost framework offers a highly accurate, fair, and explainable solution for cardiovascular disease prediction. Its combination of strong discriminative performance and transparent feature attribution provides a solid foundation for the development of AI-assisted screening tools and precision medicine systems aimed at early detection, improved clinical decision-making, and better patient outcomes.

Abbreviations

AI	Artificial intelligence
AUC	Area under the curve
CB	CatBoost
CSI	Critical success index
CV	Cross-validation
CVD	Cardiovascular disease
CVDs	Cardiovascular diseases
ET	Extra trees
FN	False negative
FP	False positive
GB	Gradient boosting
HDC	Heart disease classification
MAE	Mean absolute error
ML	Machine learning
MSE	Mean squared error
NPV	Negative predictive value
PPV	Positive predictive value
PR-AUC	Precision–recall area under the curve
RMSE	Root mean squared error
ROC	Receiver operating characteristic
ROC-AUC	Receiver operating characteristic area under the curve
SMOTE	Synthetic minority over-sampling technique
TN	True negative
TP	True positive
TPR	True positive rate
XAI	Explainable artificial intelligence
XGB	Extreme gradient boosting

Acknowledgements

The authors would like to extend their sincere appreciation to the Ongoing Research Funding program (ORF-2026-301), King Saud University, Riyadh, Saudi Arabia.

Author contributions

Redwana Khawla: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization Md. Alamin Talukder: Conceptualization, Methodology, Supervision, Software, Validation, Formal analysis, Investigation, Project administration, Visualization, Writing - Original Draft, Writing - Review & Editing Md. Abu Layek: Supervision, Software, Validation, Formal analysis, Investigation, Project administration, Visualization Utpal Kanti Das, Ammar Alazab: Methodology, Data Curation, Visualization, Validation. Mohsin Kazi: Methodology, Resource, Formal Analysis, Visualization, Validation, Writing-review & editing.

Funding

Open Access funding enabled and organized by CAUL and its Member Institutions

Data availability

The selected datasets are sourced from free and open-access sources such as HDC: <https://www.kaggle.com/datasets/bharath011/heart-disease-classification-dataset> and CD: <https://www.kaggle.com/datasets/jocelyndumlao/cardiovascular-disease-dataset>.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 26 June 2025 / Accepted: 2 February 2026

Published online: 16 February 2026

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